

SIMULATE

$$u_{t+1} = G_\theta(u_t, a_t)$$

WHAT IT DOES

Fourier layers $v_{\ell+1} = \sigma(Wv_\ell + \mathcal{F}^{-1}[R_\ell \cdot \mathcal{F}[v_\ell]])$ with modes truncated at $k_{\max}$ of 12.

Trained on $\|G_\theta - u'\|^2$ with rollout and physics residual terms; spectral normalisation bounds L_G .

A U-Net path carries sharp fronts, combined through a gate that is free to take any value.

WHAT HELD UP

0.3162 on the NDWS test split, which is
111 percent of the published benchmark.

Resolution invariant to three decimals
from 64 to 128 squared. On PDEBench
shallow water, 0.0018 at one step.

WHAT DID NOT

Proposition 1 is vacuous in its infinite
horizon form: 422 against a measured
0.10. Report the finite horizon form and
say that it is still 100 times loose.
The physics loss buys speed, not accuracy.

*The learned gate over the spectral path, initialised at zero and free to grow, settles at 0.12 on every seed and split.
Offered the Fourier operator on sharp fire fronts the model declines it, because a truncated spectrum cannot
represent a front.*